

4 Many people swim as part of their regular exercise.

(a) The table gives some information about the effect of the intensity of exercise on heart rate.

Intensity of exercise / a.u.	Heart rate / bpm
4	118
6	120
8	138
10	150
12	155
14	160

Describe the relationship between the intensity of exercise and heart rate.

(2)

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(b) Regular swimming affects resting heart rate and cardiac output.

$$\text{cardiac output} = \text{stroke volume} \times \text{heart rate}$$

The table shows the mean resting cardiac output and mean resting stroke volume for a group of regular swimmers and a control group.

Group	Mean resting cardiac output / $\text{dm}^3 \text{ min}^{-1}$	Mean resting stroke volume / cm^3
Regular swimmers	4.43	74.40
Control	4.21	58.40

Using the data in the table, calculate the difference in resting heart rate between these two groups.

Give your answer to 2 decimal places with appropriate units.

(3)

Answer

* (c) Explain the role of the heart in responding to regular exercise.

Use the information in **both** tables and your own knowledge to support your answer.

(6)

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(Total for Question 4 = 11 marks)